

CCTNS Datasets Comparative Analysis Report

Structural, Field-Level & Relational Evaluation of Criminal Justice Datasets

Target Datasets: Accused, Accused Details, FIR, Court

Scope: CCTNS Statewide Database

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1. Executive Summary

This report delivers an end-to-end architectural and field-level comparison of the four primary datasets extracted from the CCTNS (Crime and Criminal Tracking Network & Systems) backend: **Accused**, **Accused Details**, **FIR Data**, and **Court Records**. The evaluation reveals that while all four datasets converge on a single master universe of **4,596 distinct FIRs**, the two accused-centric datasets fulfill fundamentally different operational roles: **Accused Details** serves as a lightweight statewide index with primary person identifiers, whereas **Accused** functions as an exhaustive, multi-dimensional investigative dossier.

2. Overview of the Four Datasets

Dataset File	Source Variable	Records	Fields	Operational Purpose & Content
accused_records.json	ACCUSED_API_URL	938 (2014-15)	140	Deep Dossier: Biometrics, KYC, seizures, 12 interrogation relatives, & FIR narrative.
accused_details_records.json	ACCUSED_DETAILS_API_URL	20,197 (All years)	22	Master Roster: Statewide directory with CCTNS unique PERSON_CODE.
fir_records.json	FIR_API_URL	7,305 (4,596 FIRs)	11	FIR Master: Police station metadata, registration dates, and full text complaint.
court_records.json	COURT_API_URL	7,660 (4,596 FIRs)	16	Judicial Tracking: Court name, CC number, chargesheet dates, and disposal outcome.

3. Deep Dive: Accused vs. Accused Details

Although both files capture information regarding individuals accused in criminal cases, they diverge significantly in scope, structure, and database origin.

A. Commonalities & Identical Concepts

- Exact Matching Field Names (13 Fields):** FIR_NO, FIR_REG_NUM, REG_DT, FIR_STATUS, ACCUSED_NAME, AGE, GENDER, CASTE, NATIONALITY, FATHER_NAME, MOBILE_1, TELEPHONE_RESIDENCE, ARREST_SURRENDER_DT.
- Semantically Equivalent Fields with Varied Naming:**

Data Concept	accused_records.json	accused_details_records.json
Police District / Unit	DISTRICT	UNIT
Police Station	PS	PS_NAME
Registration Year	YEAR	REG_YEAR
Section of Law	ACT_SEC	SECTION_OF_LAW
Occupation	ACCUSED_OCCUPATION	OCCUPATION
Current Address	PRESENT_ADDRESS	ACCUSED_PRESENT_ADDRESS
Permanent Address	PERMANENT_ADDRESS	ACCUSED_PERMANENT_ADDRESS

B. Structural Differences & Unique Attributes

Category	accused_records.json (Accused)	accused_details_records.json (Accused Details)
Column Depth	140 Columns: Exhaustive investigative fields.	22 Columns: Lean, normalized profile.
Primary Key	Composite: (FIR_REG_NUM + ACCUSED_NAME)	Unique CCTNS ID: PERSON_CODE (18 digits)
Drug Seizures	11 fields (DRUG_TYPE, WEIGHT_GM, value, etc.)	None (not captured)
KYC & Identity	15 fields (Aadhar, PAN, Voter, Passport, Gas, etc.)	None (phone & father name only)
Biometrics	26 physical attributes (height, teeth, eyes, scars)	None (standard demographics only)
Interrogation Tree	48 fields (12 family/friend contacts & jobs)	None (father name only)
API Performance	Oracle PL/SQL procedure; requires monthly batching.	Standard REST API; fetched 20,197 records in 1 call.

4. Relational Data Architecture & Linkage

The CCTNS database utilizes a centralized star-schema model centered on the FIR. All four JSON files can be joined cleanly without data loss using standard relational keys:

Source Dataset	Foreign Key Link	Target Dataset	Relationship Type
fir_records.json	FIR_REG_NUM, FIR_NO	accused_details_records.json	1 : N (One FIR has Many Accused)
accused_details_records.json	FIR_REG_NUM, ACCUSED_NAME	accused_records.json	1 : 1 (Accused details to Deep Dossier)
fir_records.json	FIR_REG_NUM, FIR_NO	court_records.json	1 : N (One FIR has Disposal/Hearings)

5. Year-by-Year Record Distribution

Analysis of timestamps (REG_YEAR / REG_DT) across all four datasets reveals coverage spanning **28 distinct years (1991 to 2022)**. Peak enforcement activity is recorded in **2021** (4,102 accused, 1,353 FIRs) and **2022** (3,396 accused, 908 FIRs).

Year	Accused (2014-15)	Accused Details	FIR Data	Court Records
1991 - 2000 (Historical)	0	62	22	23
2001 - 2005	0	1,650	755	795
2006	0	611	331	348
2007	0	417	224	233
2008	0	380	123	125
2009	0	460	163	173
2010	0	540	164	172
2011	0	550	171	189
2012	0	571	201	214
2013	0	571	227	255
2014	382	428	213	221
2015	556	443	202	216
2016	0	705	265	286
2017	0	1,024	404	425
2018	0	803	307	322
2019	0	1,361	464	484
2020	0	1,622	556	577
2021 (Peak)	0	4,102	1,353	1,401
2022	0	3,396	908	935
Total Records	938	20,197	7,305	7,660

6. Recommended Data Engineering Strategy

- **For Statewide Roster & Multi-Year Trends:** Use `accused_details_records.json`. It covers all 28 years (1991–2022) with 20,197 records and the universal `PERSON_CODE` without database buffer constraints.
- **For Criminal Intelligence & Dossier Analysis:** Query `accused_records.json`. It provides unmatched depth (seized contraband, biometric attributes, KYC proofs, and 12-relative family networks).
- **For Multi-Year Date Queries on Accused:** Always execute queries through `getAccused.js` with its built-in adaptive monthly batching to prevent Oracle `ORA-06502` buffer overflow errors.
- **For Court Tracking:** Join `court_records.json` on `FIR_REG_NUM` to append Calendar Case numbers (`COURT_CASE_NUM`) and disposal outcomes (Acquittal/Conviction) across any year.